Improve and Challenge

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Before you start

Week 28: comparisons need fixed requests and criteria.

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Repair a specific observed failure. Try the recall before opening notes.

Your learning target

I can repair a specific observed failure.

I can check that earlier behavior still works.

I can keep a truthful change and failure record.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Test Diverse Users. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Project Problem and Success Test. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from **Evidence** and Sources. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 35: Improve and Challenge

A **controlled improvement** changes one design choice and reruns the same tests.



THINKER MISSION

Compare **before/after** scores and explain the **tradeoff**.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Why rerun old passing cases after a fix?

Readiness for Testing functions and interfaces

Why can testing only valid_item miss a pipeline bug?

My readiness explanation

This week's end goal

By the end, I can make a focused repair and challenge it with **regression** and order-invariance tests.

Notice the evidence

> Investigate the supplied Improve and Challenge case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use improve and challenge to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

Write or draw your reasoning and evidence.

My **prediction** and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

A fictional chooser wrongly excludes an activity at the exact time limit because it uses minutes < limit. The requirement says at most, so equality must qualify.

Repair that condition, then test below, at and above the boundary plus earlier tie and invalid-**input** behavior.

Apply the connected idea: Testing functions and interfaces

Test the chooser at two scales. A **unit test** checks one validator or score calculation. An **integration test** runs validation through selection and reporting.

Write expected outcomes from the specification, not by calling the same function twice.

STUDENT LESSON · 3 OF 18

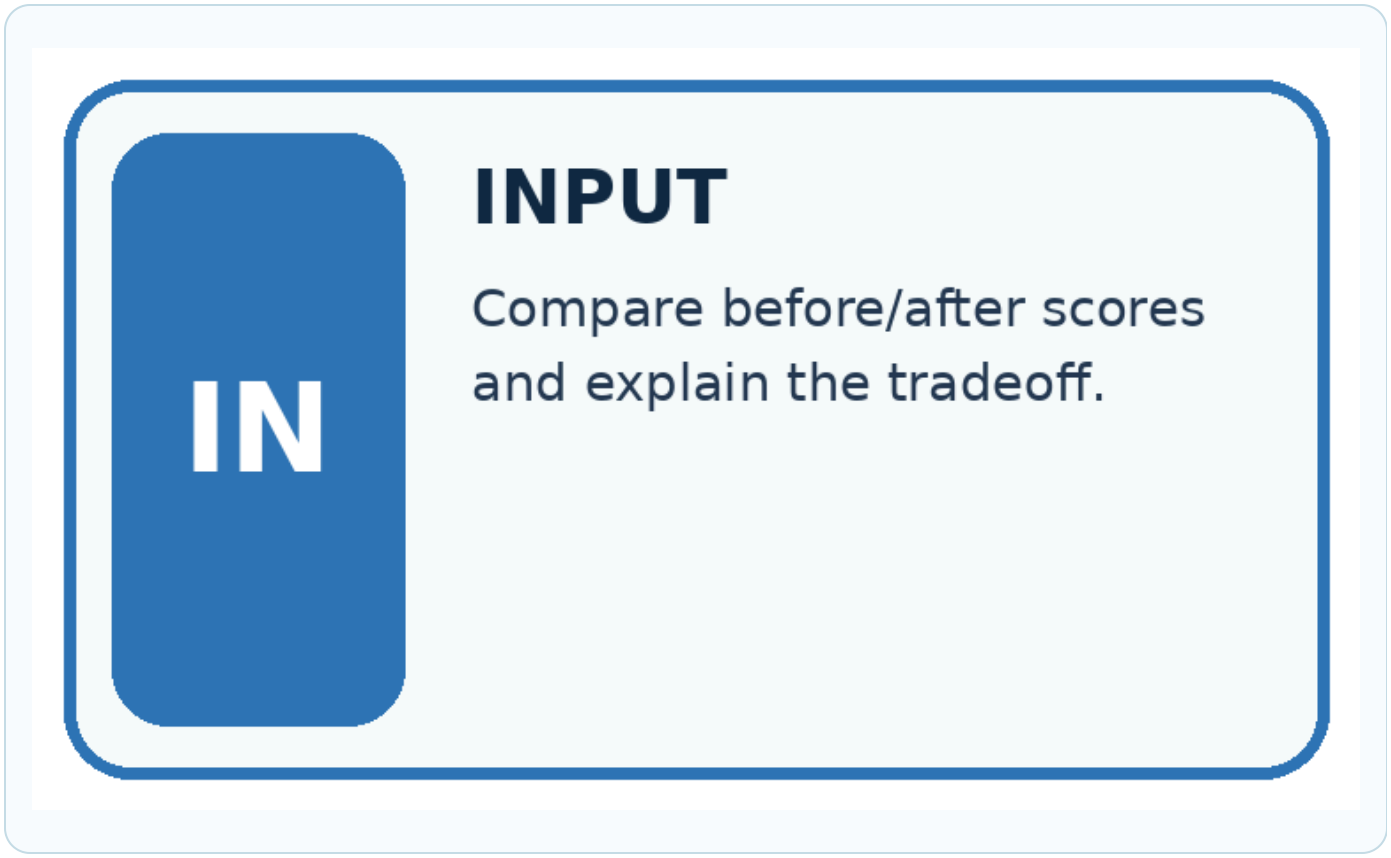
See how it works

> Follow the mechanism for Improve and Challenge. Connect each labelled input, deciding step and result.

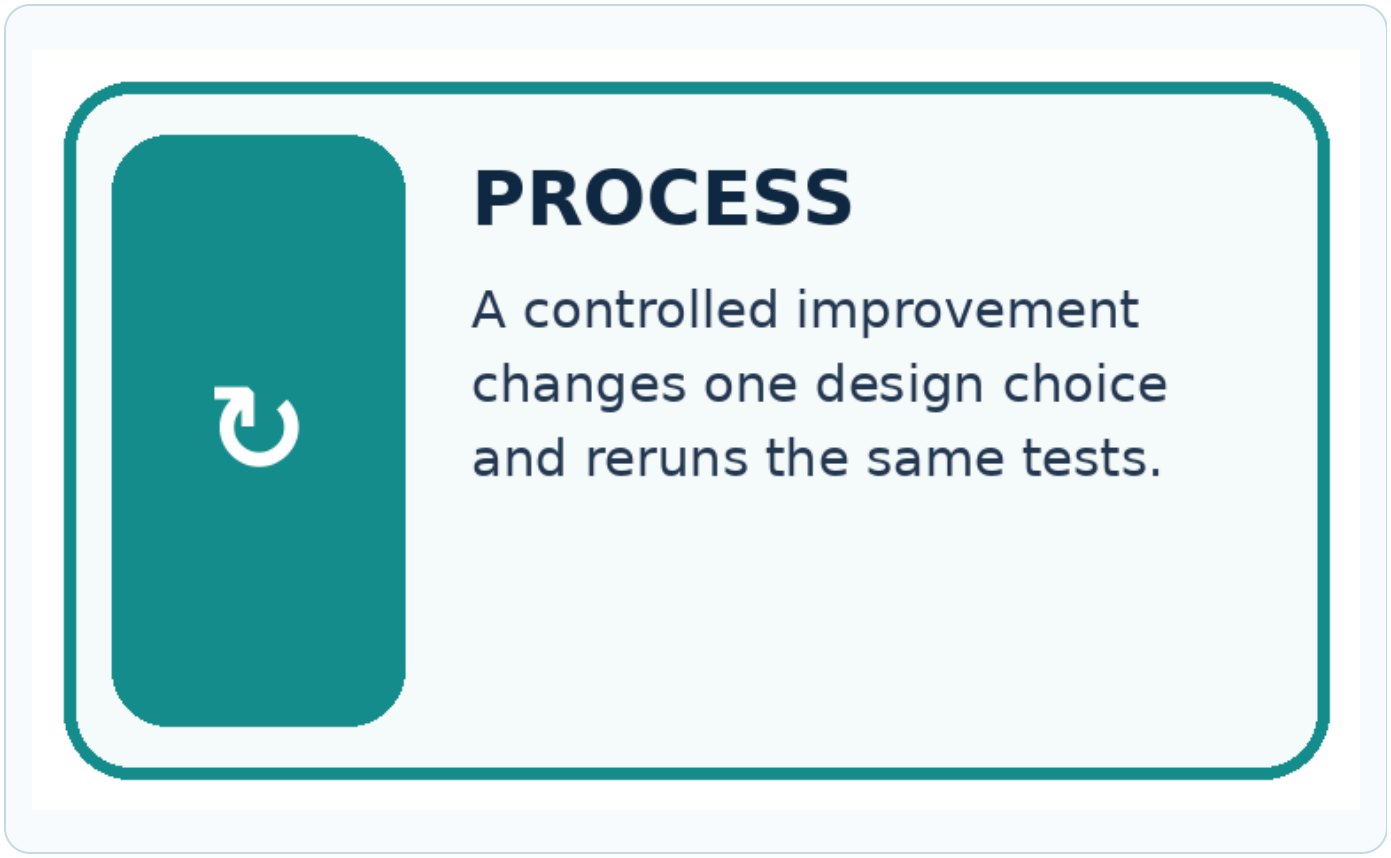
VISUAL MECHANISM

Input → Process → Output

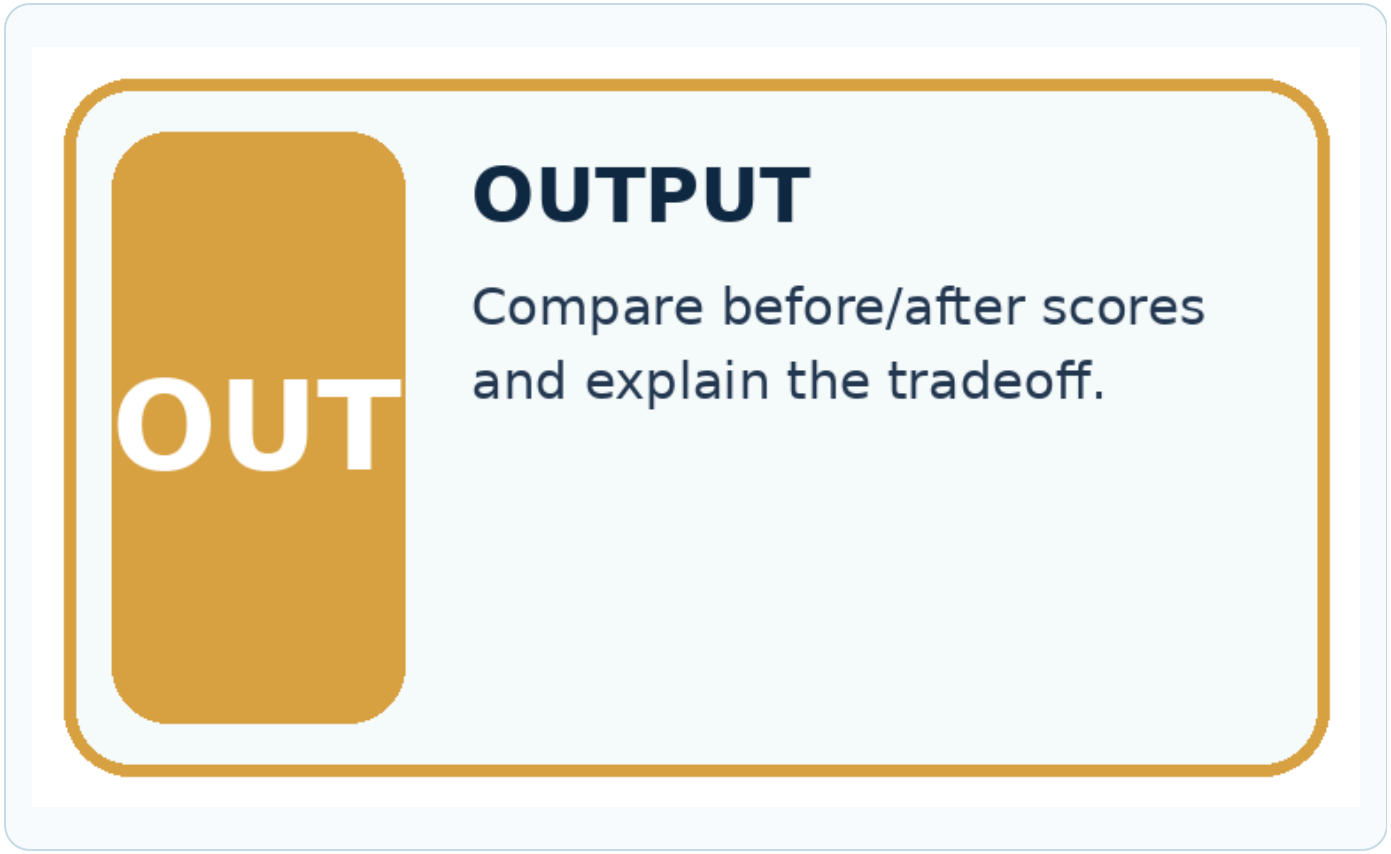
Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Compare before/after scores and explain the tradeoff.

↓


PROCESS
A controlled improvement changes one design choice and reruns the same tests.

↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

```
def eligible(minutes, limit):
    return minutes <= limit

for minutes, expected in [(9, True), (10, True), (11, False)]:
    actual = eligible(minutes, 10)
    assert actual == expected
    print(minutes, actual)
```

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
Before fix	10 < 10 is False	Violates at most rule
After fix	10 <= 10 is True	Boundary accepted
Neighbor check	11 <= 10 is False	Over-limit still excluded

Read each row from left to right and connect the evidence to the stated rule.

The repaired predicate prints 9 True, 10 True and 11 False. It assumes validated numeric durations; it is not a replacement for the schema validator.

A focused change is easier to explain and evaluate than unrelated edits made at the same time.

Repair the rule, then keep the evidence

At most includes equality. For a limit of 10, the durations 9 and 10 qualify while 11 does not. The strict comparison minutes < limit wrongly rejects 10; changing it to minutes <= limit repairs that boundary.

Check below, at and above the boundary, then rerun earlier tie and invalid-**input** cases. Keep the original failure record so another person can understand what changed and which **evidence** supports the repair.

Apply the connected idea: Testing functions and interfaces

```
def score(item):
    return 2 * item["fun"] + item["learning"]

cases = [( {"fun": 0, "learning": 0}, 0),
         ( {"fun": 3, "learning": 2}, 8),
         ( {"fun": 5, "learning": 5}, 15)]
for item, expected in cases:
```

```
    assert score(item) == expected
print("3 score cases passed")
```

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
Unit	Score 3 and 2	Expect 8
Integration	Invalid high-score row	Must never win
Report	No eligible rows	Explain no suggestion

Read each row from left to right and connect the evidence to the stated rule.

The code prints its success message only if all assertions pass. It checks the scoring component but says nothing about duration filtering, malformed records, ties or reporting. Add focused cases for those requirements.

Use the production functions from earlier weeks rather than rewriting a second copy just for tests.

Test the connections as well as the pieces

A score **unit test** asks whether the scoring function calculates the specified number. An **integration test** asks whether validation, filtering, scoring and reporting work together.

A correct score does not establish that invalid records were rejected or that a tie was resolved correctly. Predict the full report from the **input** records before running.

Check its counts and reasons as well as its selected name, and repeat a tie case with reversed **input** order.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

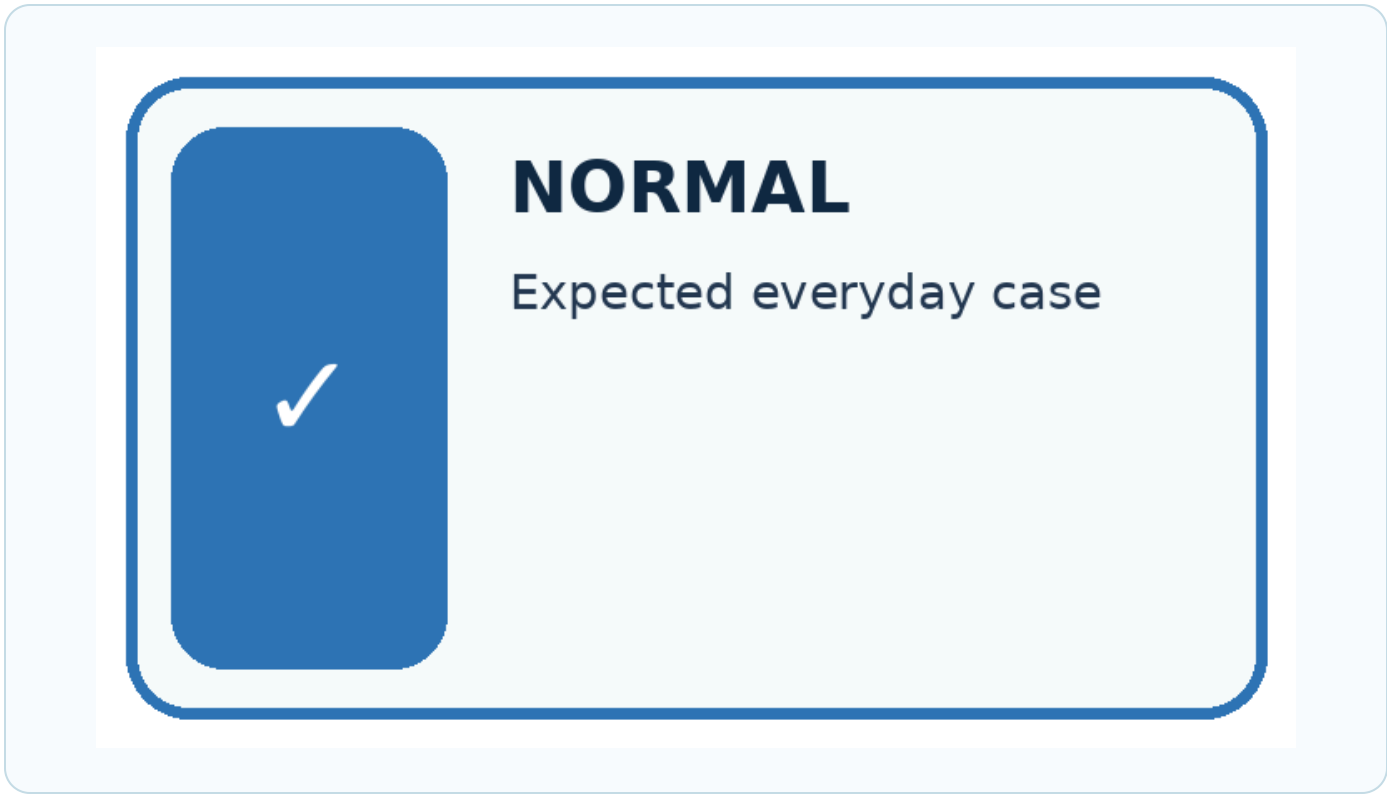
> Compare the supplied cases for Improve and Challenge. Identify the detail that changes the conclusion.

The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

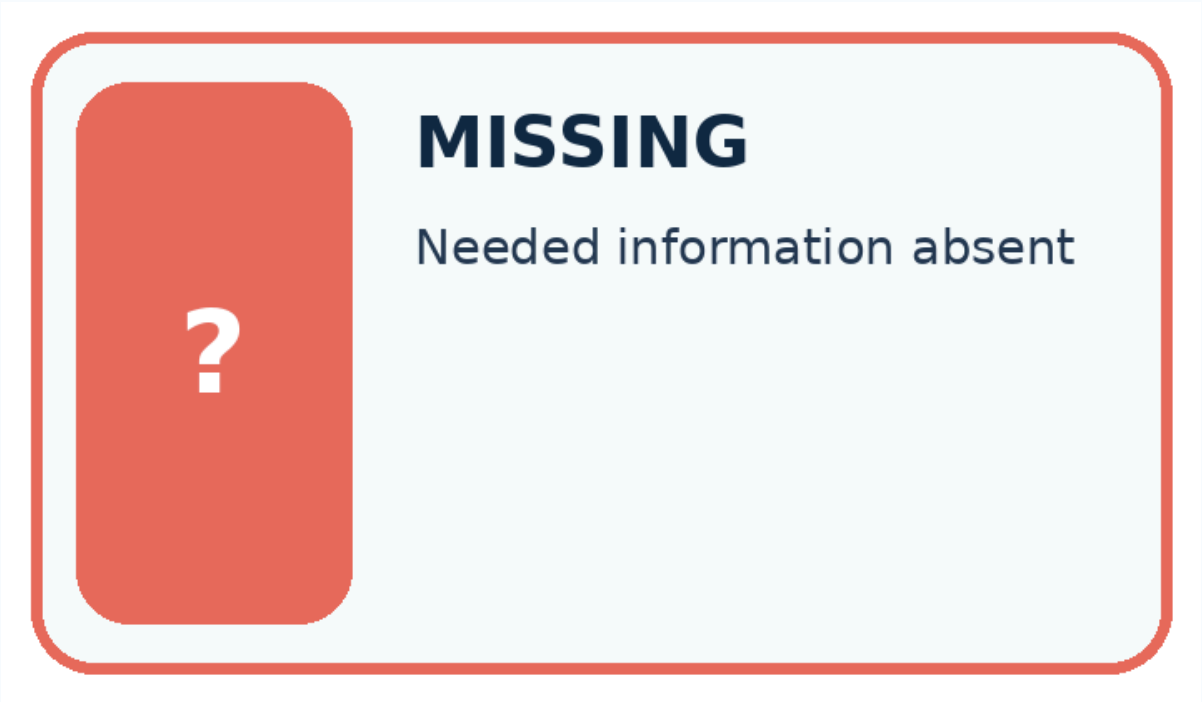
A system is not understood until it survives more than one comfortable example.


NORMAL
Expected everyday case


BOUNDARY
Near a decision edge

The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: An alternative patch simply increases the limit to 101.

Predicted result

Prediction: C is wrongly admitted as well as A.

Why the result follows

Reason: Changing the maximum alters the contract instead of repairing its comparison.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Improve and Challenge to explain a concrete decision, rather than only memorising definitions.

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

1

CONTROLLED IMPRC

one planned change tied to evidence

2

BEFORE/AFTER

results from the same test before and after





Word	My own example	Not this / boundary
controlled improvement	A revision whose effect is checked under comparable conditions rather than assumed from a changed result.	
before/after	Results recorded prior to and following a change; other changes can affect the comparison.	

Word	My own example	Not this / boundary
tradeoff	A situation where improving one desired property can make another harder to achieve.	
regression	In testing, a new failure introduced by a change; in modelling, prediction of a numeric target. Use the meaning indicated by the lesson.	

Words used in the closer explanation

Word	Meaning
Regression suite	Earlier cases rerun after a change
Change log	A record of what changed and why
Root cause	The underlying reason for a failure
Residual risk	A limitation or possible failure remaining after a repair

Terms for Testing functions and interfaces

Term	Meaning
Unit test	A check of one small component
Integration test	A check of connected components
Fixture	A controlled input used in a test
Oracle	An independent source of the expected result

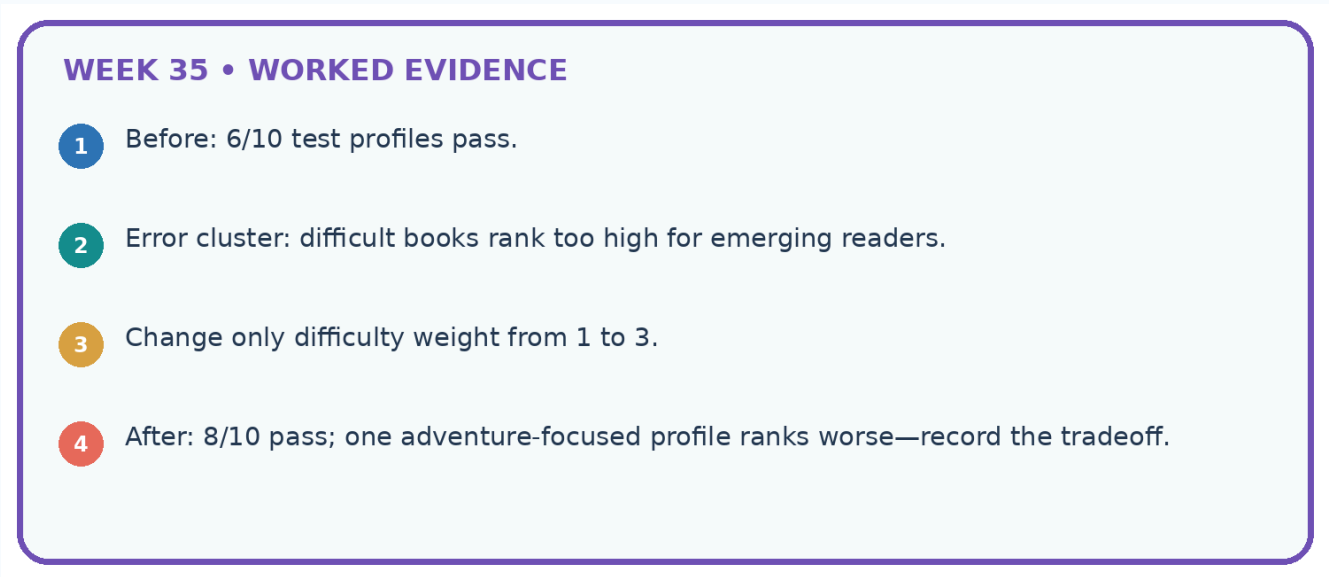
Follow a worked example

> Follow the worked trace for Improve and Challenge, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Before: 6/10 test profiles pass.

2. Error cluster: difficult books rank too high for emerging readers.

3. Change only difficulty weight from 1 to 3.

4. After: 8/10 pass; one adventure-focused profile ranks worse—record the **tradeoff**.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Would changing the condition to minutes <= limit + 1 correctly repair the boundary? Explain with a revealing input.

Apply the connected idea: Testing functions and interfaces

> **A test computes expected = score(item) then compares score(item) to expected. Why is it weak?**

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Improve and Challenge

A fictional recommender excludes books at an intended including equality page limit.

The facts we will use

> **Supplied facts**
>
> Contract: pages<=100; highest interest; ties choose smaller ID. Old code uses pages<100.

> **Supplied facts**
>
> Rows A pages100 interest5; B pages90 interest3; C pages101 interest5.

AI ACADEMY / AI-2/35

Improve and Challenge

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Identify cause

A wrongly excluded at equality, so old code selects B.

2. Focused repair

Replace <100 with <=100 without changing ranking or type checks.

3. Expected revised result

A becomes eligible and wins; C remains excluded.

4. Challenge repair

Rerun 99,100,101; invalid pages; tied scores; reversed row order.

CHANGE ONE THING

An alternative patch simply increases the limit to 101.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Identify cause	A wrongly excluded at equality, so old code selects B.
2. Focused repair	Replace <100 with <=100 without changing ranking or type checks.
3. Expected revised result	A becomes eligible and wins; C remains excluded.
4. Challenge repair	Rerun 99,100,101; invalid pages; tied scores; reversed row order.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A focused change makes the claim about what caused the error testable: the equality comparison rule caused this specific failure. **Regression** tests check surrounding behavior that should remain unchanged.

A tie case and reversed **input** order are necessary because a correct boundary does not ensure deterministic selection. Log the old failure and revised replay separately.

What this example does not establish

Keep the stated limit

Passing a small **regression suite** does not prove all behavior unchanged; record the suite's actual coverage.

Keep the stated limit

Read the labelled picture: Why are tests at both 100 and 101 needed?

Keep the stated limit

Change one thing: An alternative patch simply increases the limit to 101. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Improve and Challenge claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: If the total score improves, no old behavior needs retesting. Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

A clean **regression** run does not erase the earlier failure. Keep **before/after evidence** and remaining limitations. If a fix was informed by reserved evaluation cases, report that contamination and obtain new independent cases.

> **Record the bug, change, evidence and one remaining limit without claiming all bugs are gone.**

Apply the connected idea: Testing functions and interfaces

Test results establish only the executed cases on the tested version. Retain failures and version information. Use explicit expected rejection behavior for invalid data instead of assuming every exception is acceptable.

> **What expected result should an empty-input integration case specify?**

STUDENT LESSON · 8 OF 18

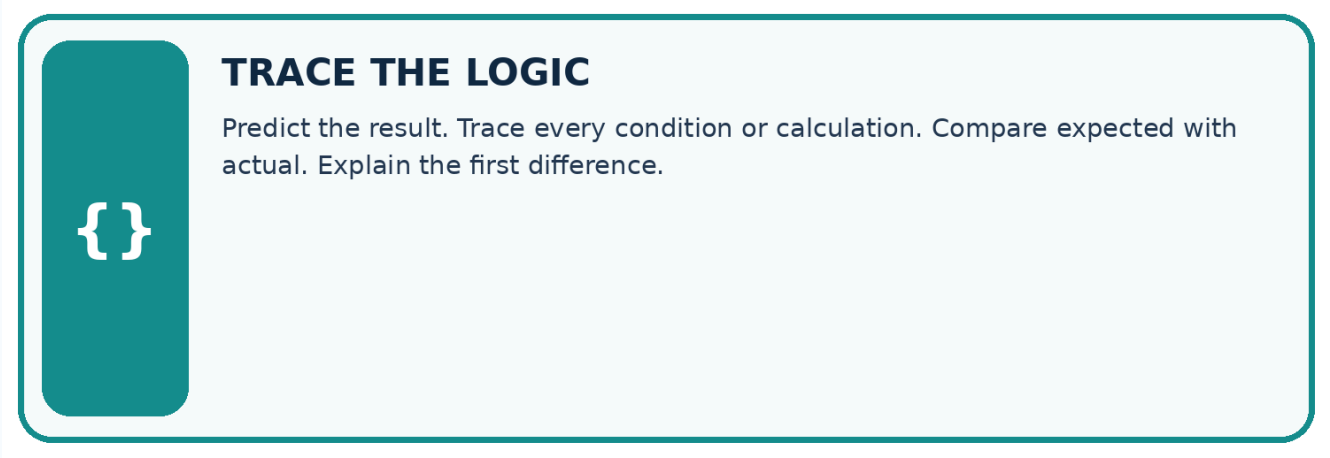
Practise together

> Practise Improve and Challenge with support, then explain the deciding step in your own words.

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

> Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Changing the expected boundary result to False would make the test pass while violating the requirement. Keep the requirement and independently calculated expectation; repair the implementation.

If requirements truly change, document that as a policy change.

Apply the connected idea: Testing functions and interfaces

A test suite that only checks the winner name can miss wrong rejection counts. Check the full expected report: supplied, accepted, rejected, eligible, selected and reason.

Avoid relying on dictionary display order when the contract concerns values.

Practise with less help: Test neighbors

Use workbook W2 for the connected example “Improving without hiding failures”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Improving without hiding failures. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Practise the connected topic: Testing functions and interfaces

Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A successful run proves the conclusion

Explain the picture

> Why are tests at both 100 and 101 needed?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

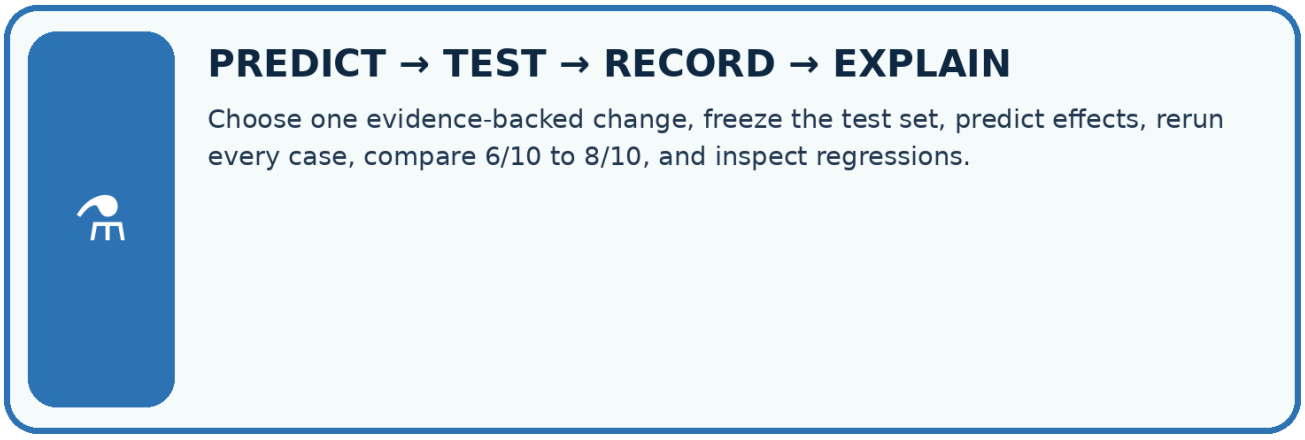
Run the investigation

> Investigate Improve and Challenge. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Choose one **evidence**-backed change, freeze the test set, predict effects, rerun every case, compare 6/10 to 8/10, and inspect **regression**s.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** boundary
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** missing
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** fresh transfer
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

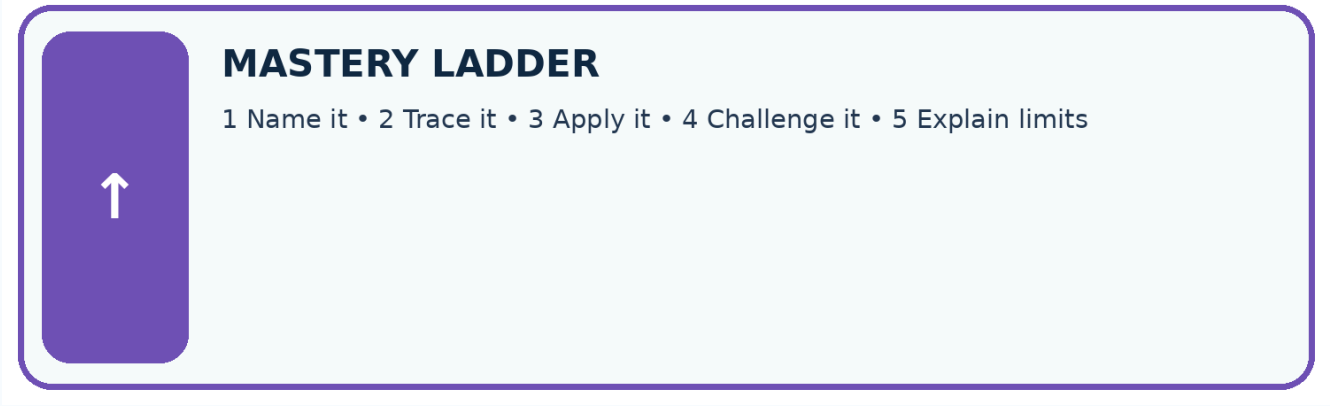
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Improve and Challenge. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

Write or draw your reasoning and evidence.

STUDENT LESSON · 11 OF 18

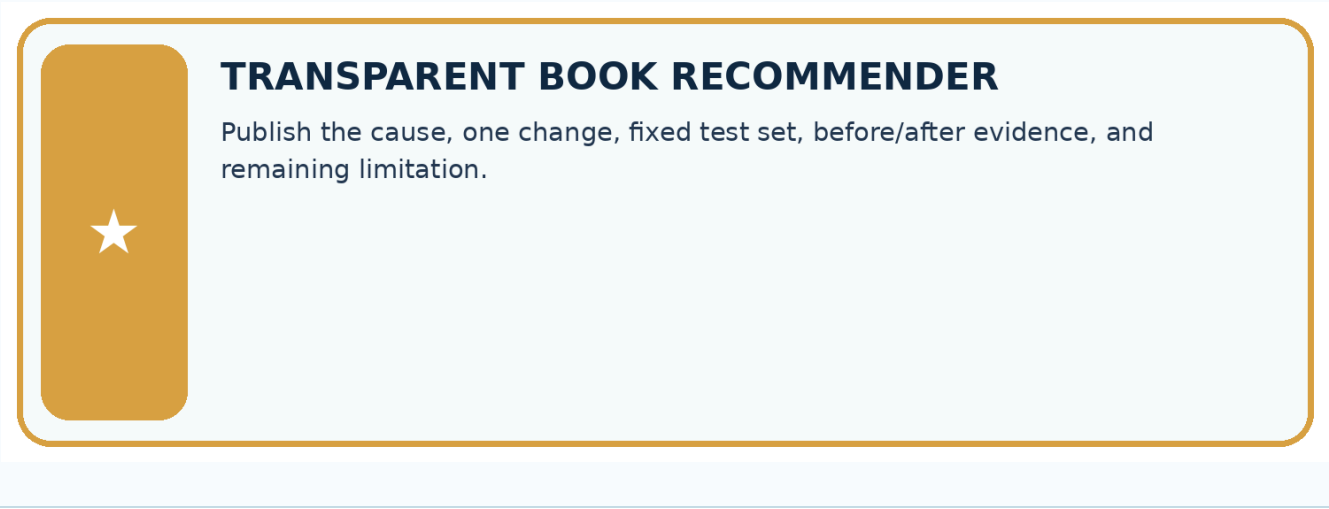
Connect your project

> Apply this week's idea to Transparent Book Recommender. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Publish the cause, one change, fixed test set, **before/after evidence**, and remaining limitation.

Trace the system

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

For each recommender revision, retain cause, code change, targeted test, **regression** results and unresolved issues.

STUDENT LESSON · 12 OF 18

Show what you know

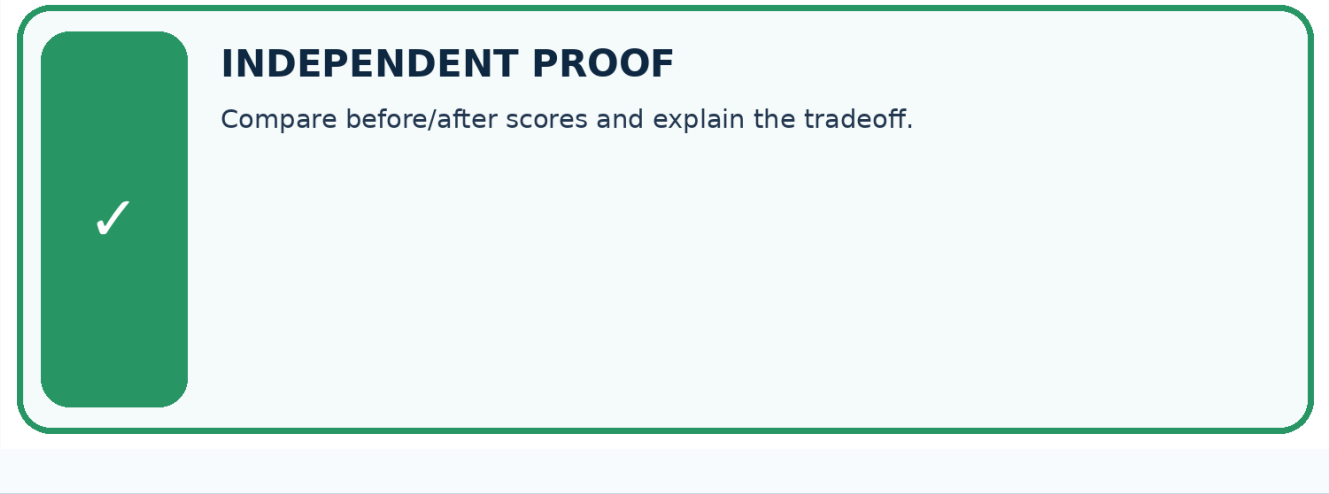
Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Compare **before/after** scores and explain the **tradeoff**.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Improve and Challenge case for supported practice. Repeated examples do not become fresh assessments.

Week 35 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A repair raises success from 7/10 to 9/10 on the same validation set.

The question

> What belongs in the improvement claim?

Worked explanation

Confirm fixed criteria, inspect remaining and introduced failures, then test the chosen version on fresh final cases.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Improve with comparable evidence

Evidence or stage	Value or result
Before	Save results and failures
Revision	Change one identified cause
Retest	Use the same development cases
Final check	Reserve fresh cases

A repair raises success from 7/10 to 9/10 on the same validation set.

> What belongs in the improvement claim?

Confirm fixed criteria, inspect remaining and introduced failures, then test the chosen version on fresh final cases.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Repair independently

Write the eligibility predicate for at most **12 minutes** and a **regression** set including 11,12,13 and 0. Add one chooser tie case and one invalid-data case to rerun.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 35 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> After changing a scoring weight, the system fixes one request but breaks two previously correct ones. What should the improvement record say?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Publish the cause, one change, fixed test set, **before/after evidence**, and remaining limitation.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

After a scoring change, four old failures pass but two previous passes fail. On the original ten cases there were six passes. Compute the new total and explain what else should be reported.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Improve and Challenge, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional recommender excludes books at an intended including equality page limit.

By the end, I can	How I will show it
Repair a specific observed failure	Explain the deciding step, show a trace or test, and state one limit.
Check that earlier behavior still works	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Keep a truthful change and failure record	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Improve and Challenge without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Repair a specific observed failure

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional recommender excludes books at an intended including equality page limit.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Rerun 99,100,101; invalid pages; tied scores; reversed row order.

1. Identify cause

2. Expected revised result

3. Challenge repair

4. Focused repair

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Challenge repair Rerun 99,100,101; invalid pages; tied scores; reversed row order.

A focused change makes the claim about what caused the error testable: the equality comparison rule caused this specific failure. **Regression** tests check surrounding behavior that should remain unchanged.

A tie case and reversed **input** order are necessary because a correct boundary does not ensure deterministic selection. Log the old failure and revised replay separately.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: An alternative patch simply increases the limit to 101. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. C is wrongly admitted as well as A.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: C is wrongly admitted as well as A. Changing the maximum alters the contract instead of repairing its comparison.

Repair a missing building block

Review Test Diverse Users (ai-2/34). Goal: Evaluate across varied cases. Short practice: Can these records establish performance for different user groups?. Return to this week and explain what you changed.

Review Project Problem and Success Test (ai-2/32). Goal: Define a small application purpose. Short practice: Write an equality test and explain which requirement it checks.. Return to this week and explain what you changed.

Review **Evidence** and Sources (ai-2/29). Goal: Explain the **mechanism** and terminology of claim support and citation evaluation.

Short practice: Why is the original answer not fully supported despite having citations on every claim?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

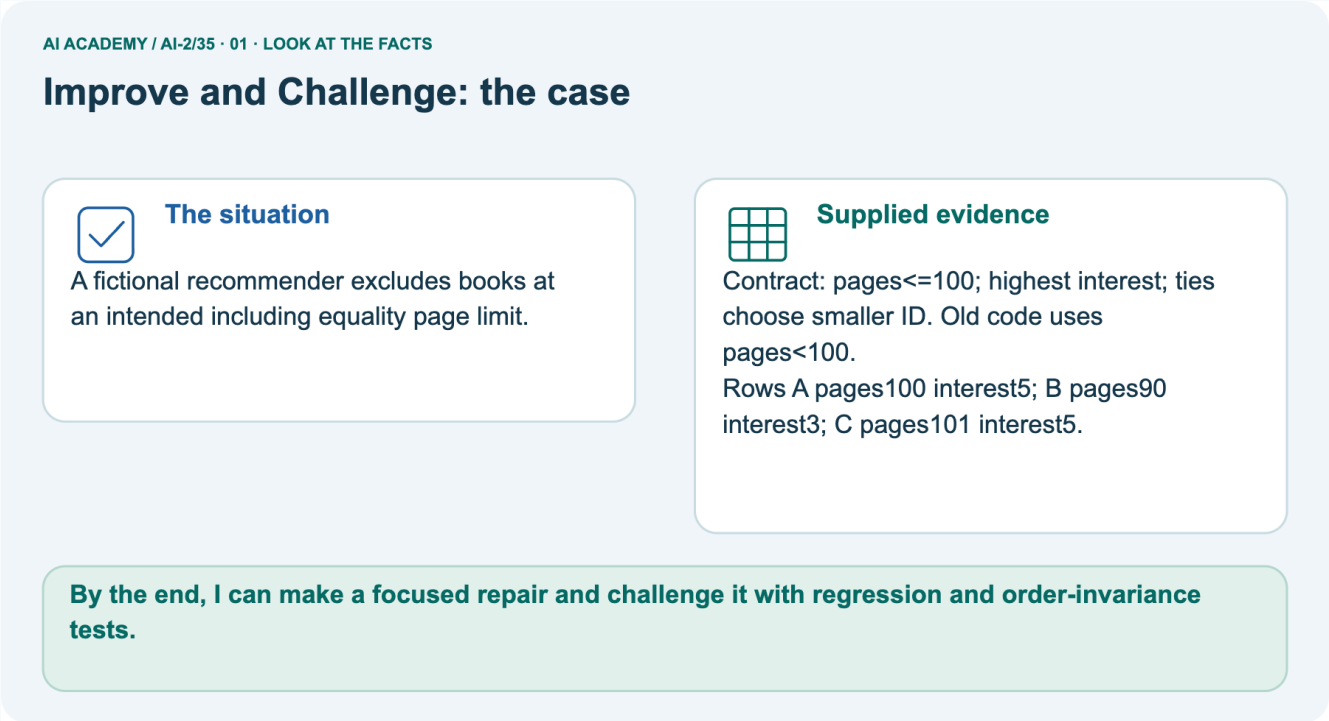
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Improve and Challenge graphic using its labels, deciding evidence and relevant limitation.

01. Improve and Challenge: the case



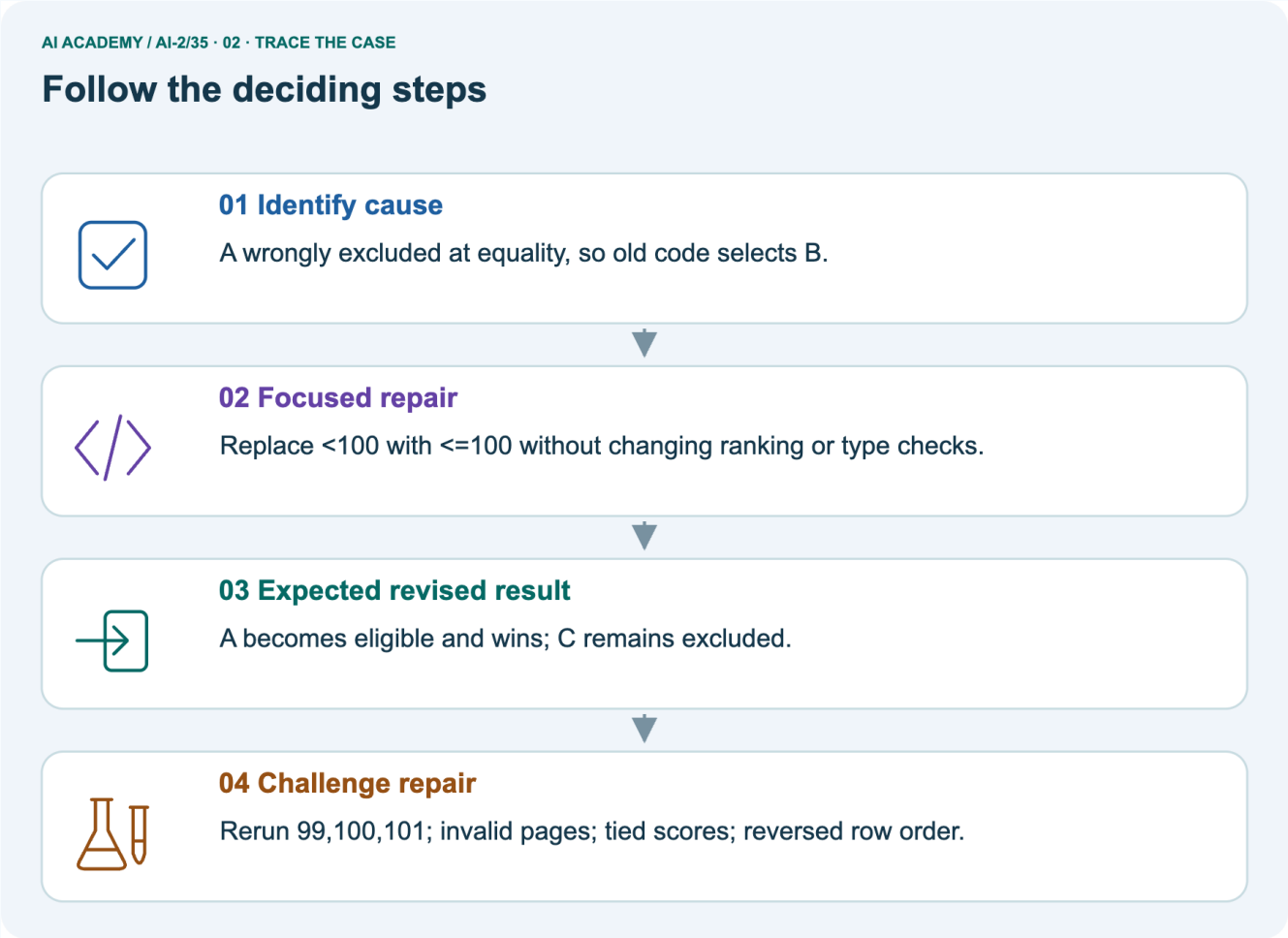
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional recommender excludes books at an intended including equality page limit. Supplied facts: Contract: pages<=100; highest interest; ties choose smaller ID. Old code uses pages<100.

Rows A pages100 interest5; B pages90 interest3; C pages101 interest5.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

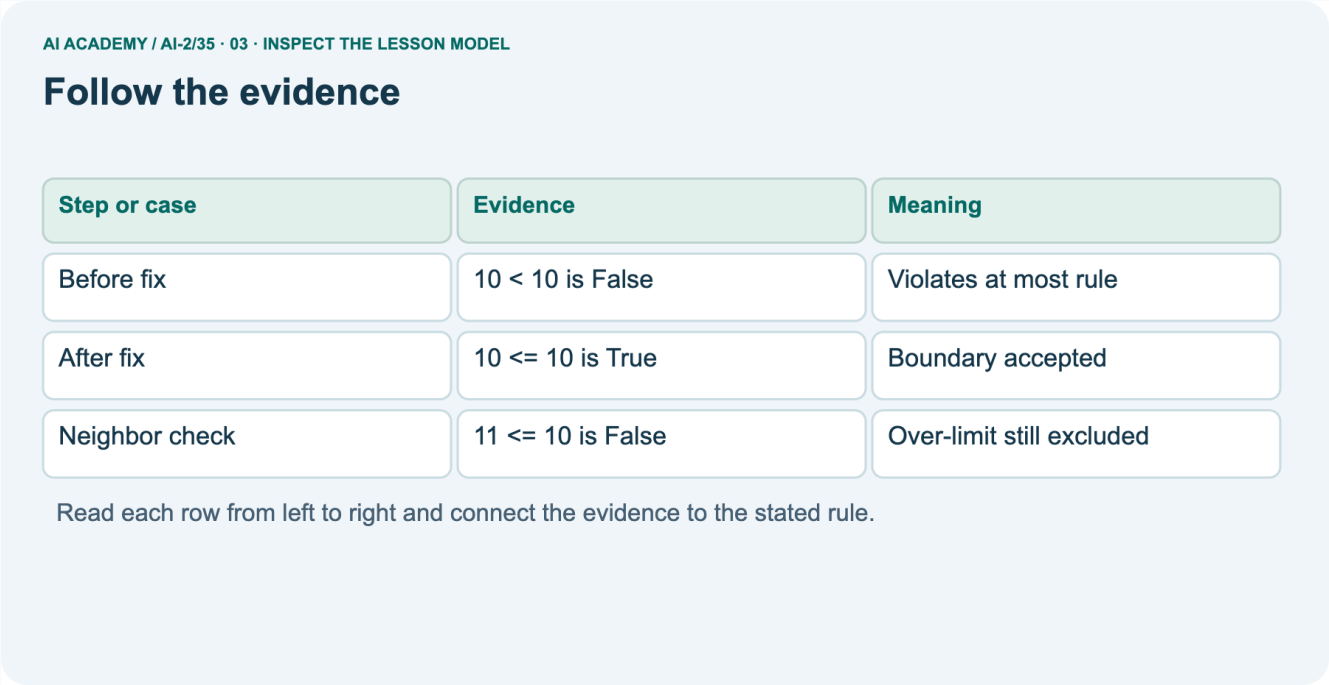


Follow the labelled picture with the matching explanation. Open the vector diagram

Identify cause: A wrongly excluded at equality, so old code selects B. → Focused repair: Replace <100 with <=100 without changing ranking or type checks. → Expected revised result: A becomes eligible and wins; C remains excluded. → Challenge repair: Rerun 99,100,101; invalid pages; tied scores; reversed row order.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Follow the evidence

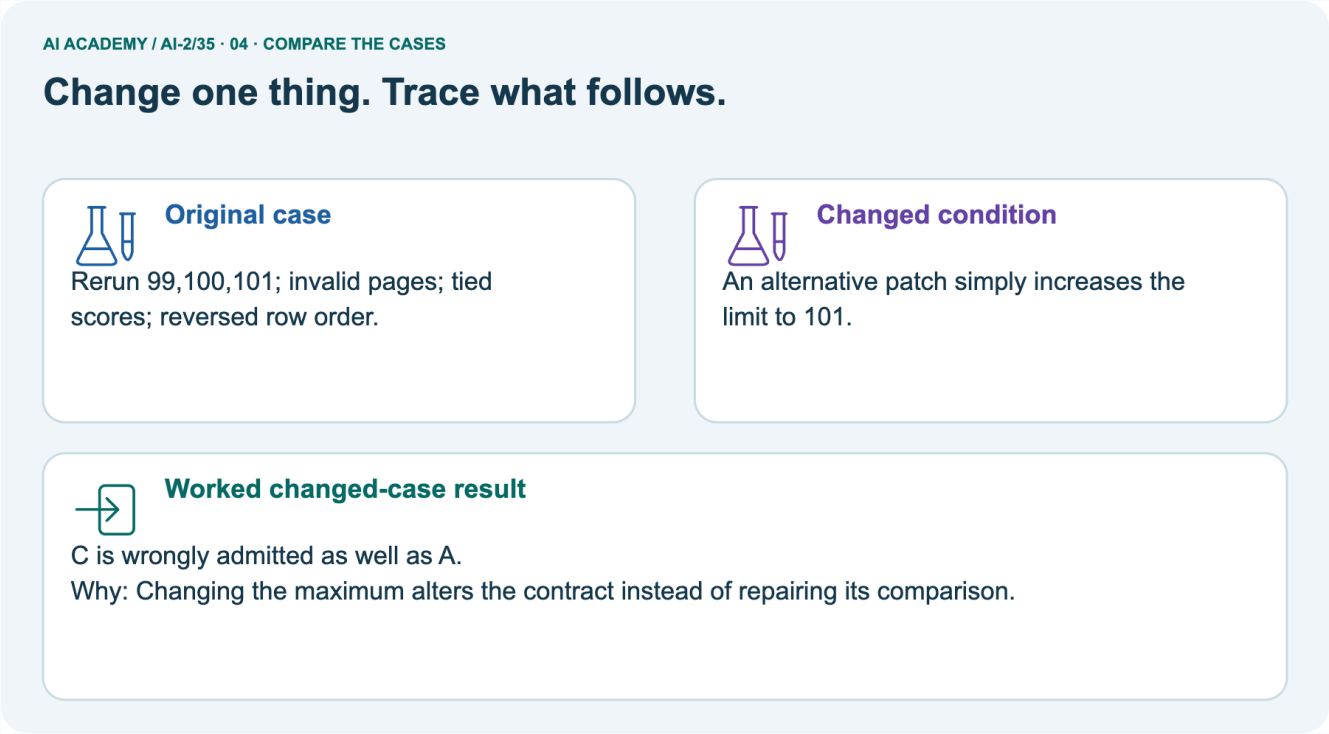


Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. Before fix | 10 < 10 is False | Violates at most rule. After fix | 10 <= 10 is True | Boundary accepted. Neighbor check | 11 <= 10 is False | Over-limit still excluded

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



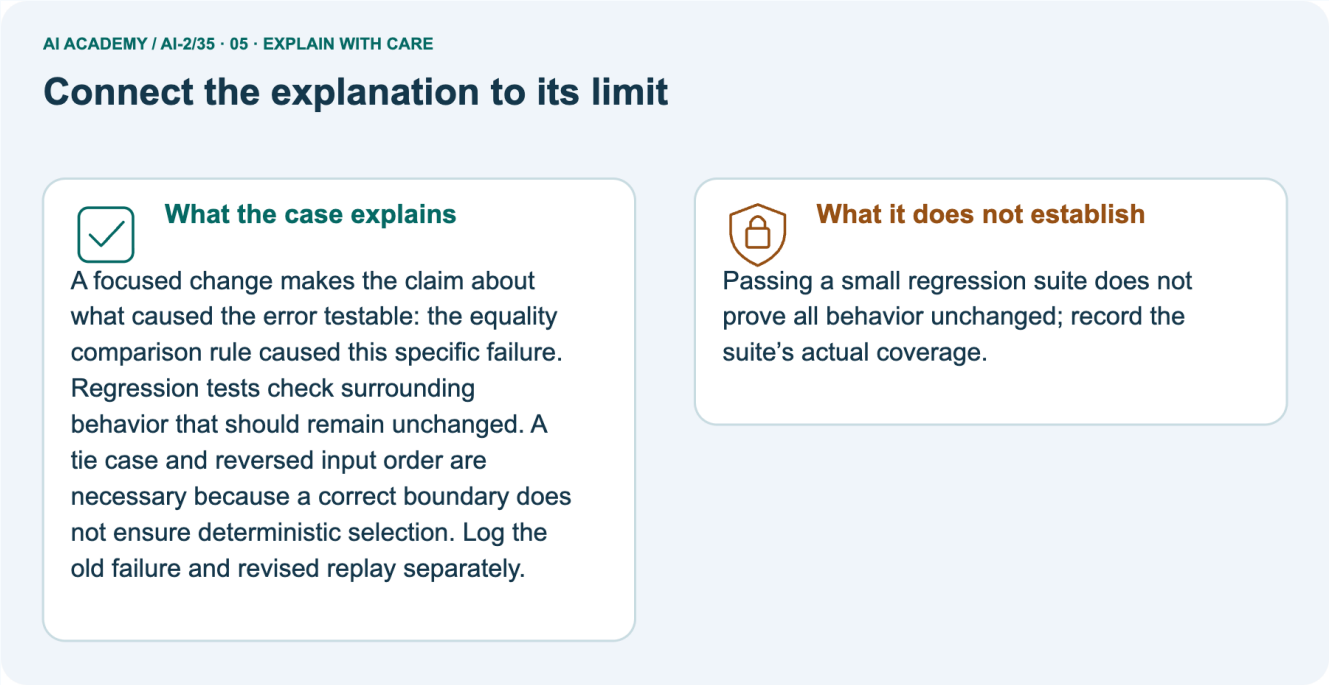
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Rerun 99,100,101; invalid pages; tied scores; reversed row order. Changed condition: An alternative patch simply increases the limit to 101. Expected result: C is wrongly admitted as well as A.

Reason: Changing the maximum alters the contract instead of repairing its comparison.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

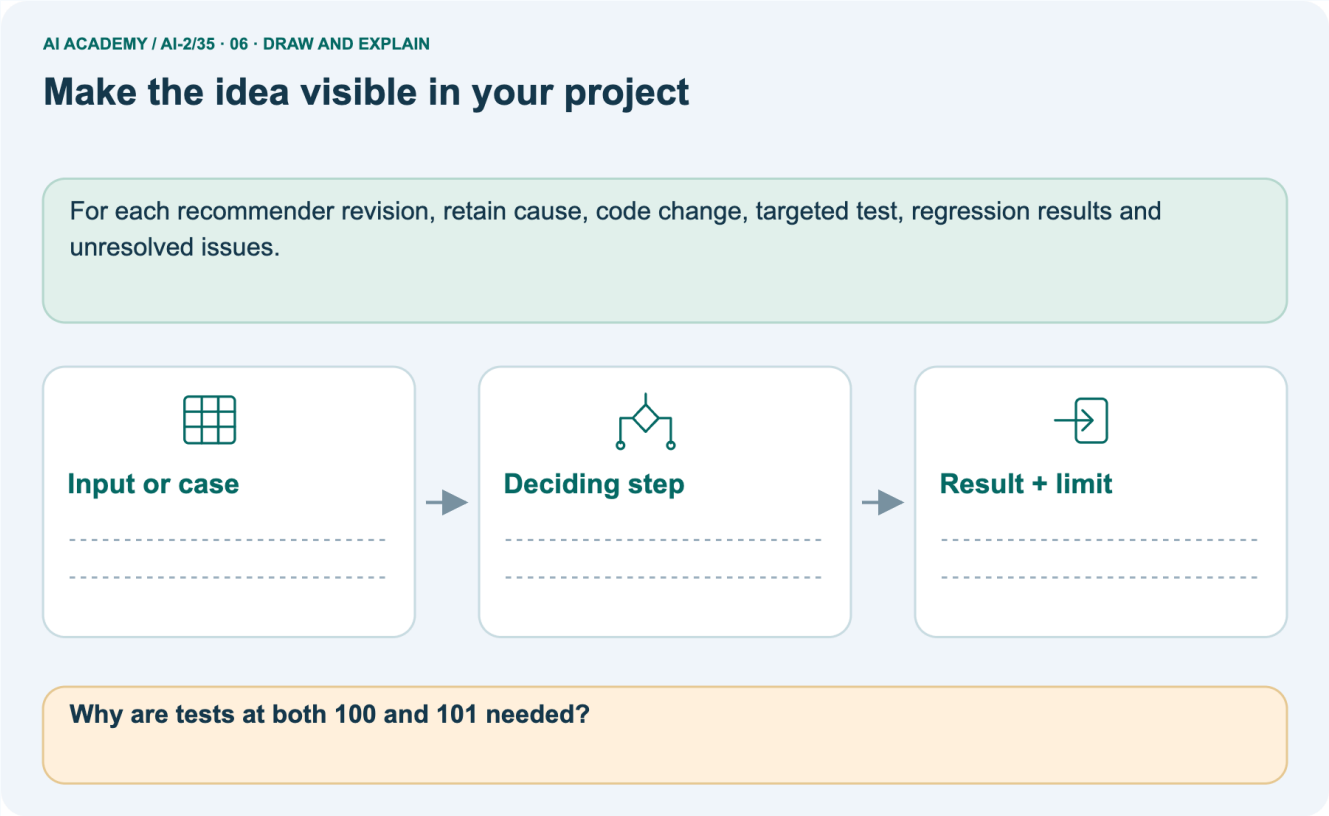
A focused change makes the claim about what caused the error testable: the equality comparison rule caused this specific failure. **Regression** tests check surrounding behavior that should remain unchanged.

A tie case and reversed **input** order are necessary because a correct boundary does not ensure deterministic selection. Log the old failure and revised replay separately.

Limit: Passing a small **regression suite** does not prove all behavior unchanged; record the suite's actual coverage.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For each recommender revision, retain cause, code change, targeted test, **regression** results and unresolved issues. Draw your **input**, deciding step, result and limit. Supported question: Why are tests at both 100 and 101 needed?

Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.